



Fig. 12: Iron anode section of prototype iron-air battery (Courtesy: Form Energy)

decreases the detrimental effect of iron passivation on the electrode by forming a conductive layer of FeS by reacting with Fe atoms. When iron oxide nanoparticles supported on tubular nanotubes of carbon particles are used as Fe electrodes, it has been seen that they yield a higher specific capacity and charging efficiency. Table 2 shows the properties of various Fe electrodes.

Cu-doped, Cu-Sn-doped, and Sn-doped nano iron active materials have been used recently as negative electrodes in alkaline batteries and showed stable, improved performance. The presence of Cu increases electrode kinetics, increases surface roughness, and reduces ohmic resistance upon cycling. Cu-doped electrodes have a performance of 360mAh g⁻¹ at 80% charging efficiency. Cu-Sn-doped nano-iron/carbon composites with CNT and LiOH addition increase the performance to 603mAh g⁻¹ with 93% charge efficiency due to uniform distribution of Cu, Sn, and Fe in the carbon network. However, cycling stability reduces due to a higher rate of hydrogen evolution at different voltages. Below 1.5V,

hydrogen evolution is minimal, and above 1.65V, it is maximal, causing rapid electrode failure. Charging at high current density increases the charging efficiency and performance while reducing hydrogen evolution due to current limitation for the hydrogen evolution reaction. Accordingly, the most optimum voltage and charge current density are selected to develop the charging algorithm. The iron anode section of a prototype iron-air battery, being fabricated at the works of Form Energy, USA, is shown in Fig. 12.

Electrolyte

For metal-air batteries, alkaline electrolytes are generally preferred over acidic electrolytes due to their higher ionic conductivity, safety in

operation, and reduced corrosiveness to electrodes. A solution of potassium hydroxide, which has good electrical conductivity and is moderately corrosive to the iron electrode, is used as the alkaline electrolyte in iron-air batteries.

During electrode reactions, metallic iron is oxidised to form Fe(OH)₂ and other oxidation products, which are sparingly soluble in the electrolyte. The low solubility helps to avoid the formation of dendrites, a harmful product. Often Na₂S, K₂S, and LiOH are used as electrolyte additives. The first two compounds work like bismuth sulphide, but over time become less effective due to non-uniformity of sulphide absorption on the electrode interface.

Recently, organosulfur compounds like n-alkanethiols were successfully used to suppress hydrogen evolution. Sulphur adsorption masks the electrocatalytic surface and remains stable over several cycles. An advantage of using organosulfur is that recycling the spent electrodes is much easier compared to those containing bismuth.

If LiOH is used, Li ions are absorbed in the iron hydroxide matrix, increasing the ionic conductivity and reaction kinetics of the iron material. The insinuated compound can be easily reduced back to iron. It has been found that the addition of potassium stannate as an electrolyte additive increases the capacity and efficiency to 400mAh g⁻¹, with

TABLE 2
PROPERTIES OF IRON ELECTRODES

Iron Material	Particle Size μm	Support Material	Additives	Discharge Specific Capacity Ahg ⁻¹	% Charging Efficiency
Commercial Fe ₃ O ₄ particles	1-3	None	None	0.12 at C/5 rate, 9mA cm ⁻²	70 at C/5 rate
Carbonyl iron	0.5-3	None	5% Bi ₂ S ₃	0.29 at C/5, at 3mA cm ⁻²	96 at C/20
Fe ₃ O ₄	0.02	Carbon nanotubes	K ₂ S	0.79 at 0.5mA cm ⁻²	84